Lithium-ion battery fires on vessels remain one of the biggest safety issues



Allianz has highlighted that fires on vessels remain one of the biggest safety issues facing the shipping industry. There have been a number of serious fire incidents in recent years where Lithium-ion (Li-ion) batteries have been reported as the source of, or contributing to fires on vessels. Allianz experts Capt. Rahul Khanna, Captain Randy Lund and Captain Anastasios Leonburg, share their thoughts on how electric vehicles may impact safety onboard.

Decarbonization and electrification are increasing the number of shipping goods that contain Li-ion batteries, from electric vehicles (EVs) to a wide range of consumer and electronic goods. The global Li-ion battery market is expected to grow by over 30% annually from 2022 to 2030, according to a report by McKinsey.

Recently, the Fremantle Highway car-carrying vessel caught fire off the Dutch coast with over 3,000 vehicles on-board enroute from Germany to Egypt. A fire on board car carrier Felicity Age in February 2022, led to the vessel sinking in the Atlantic Ocean, along with its cargo of 4,000 vehicles. Li-ion batteries were cited as being a factor in keeping the fire ablaze. The Höegh Xiamen, caught fire in June 2020 in Jacksonville, Florida, resulting in the total loss of the vessel and its cargo of 2,420 used vehicles. An improperly disconnected battery in a used vehicle led to the fire, according to the official investigation.

In light of the growing number of fires on cargo ships, a spokesperson for IMO said that it will announce new safety standards for those transporting electric vehicles in 2024. The guidelines could include specifications on how fully a battery can be charged. The IMO said that chemicals for extinguishing fires, special fire blankets, equipment such as battery-penetrating jet extinguishers and bigger gaps left between electric vehicles on ships could also become mandatory.

Allianz explains that Li-ion batteries can be carried on board ships either as a cargo themselves or as part of the equipment for the electric vehicles (EVs) they provide power for. Many of these batteries are safely transported every day but fire risks are present in both scenarios, especially if the batteries are used or defective, damaged or improperly stored, packaged, handled or labelled.

The main hazards are fire, explosion, and 'thermal runaway', a rapid self-heating fire that can cause an explosion. They can also produce irritating, corrosive or poisonous gases that cause an explosion in a confined space. The main causes of Li-ion fires are substandard manufacturing or damaged battery cells or devices, over-charging, and short circuiting. Li-ion batteries are an important source of energy and do not necessarily burn more frequently than other goods. It is only when they ignite that they are more difficult to extinguish as they can burn more ferociously and are capable of spontaneously reigniting hours or even days after they have been put out.

Most ships lack the suitable fire protection, firefighting capabilities, and detection systems to tackle such fires at sea, which has been made more difficult by the dramatic increase in ship size – container-carrying capacity has doubled in the last 20 years. We have seen many fires where malfunctioning or damaged batteries have been attributed as a contributing factor in recent years.

Allianz has long warned about the risks associated with Li-ion batteries and EVs in shipping for a number of years, first highlighting this issue in 2017. A recent report highlights a full list of loss prevention measures to consider including:

- All EVs should display clear and precise identification on the windshield detailing the battery type (e.g. Battery Electric Vehicle (BEV), Hybrid Electric Vehicle (HEV), Plug-in Hybrid Electric Vehicle (PHEV).
- EVs with low ground clearance should be clearly labelled as this can present loading and discharging challenges arising from the vessel's ramps, inner slopes, or deck appendages.
- All EVs with a Li-ion battery must have successfully passed pressure, temperature, crush, and impact tests as described in the UN Manual of Tests of Criteria – subsection 38.3 for transport of Li-ion batteries.
- All EVs must be fully functional, self-propelled, safe to drive and contain an undamaged battery system.
- There should be no charging of EVs during the passage.

- All EVs must be properly secured to prevent any shifting during transport.
- One potential idea being explored by some car carrier operators, as part of fire-preventative measures, is the use of fire-proof blankets manufactured specifically for EVs.

Allianz experts conclude that the debate about EVs in the shipping industry is ongoing, with conversations about whether there is even a need for dedicated Ro-ro vessels for EVs.

"From an insurance perspective, this is something we would like to see – purpose-built vessels for transporting EVs, designed to substantially reduce the risk of fire. We have already seen shipping companies stop transporting EVs on their ships because of the potential fire risk," says Allianz.


CM-891DM

